

Syamchand Chilaka
Data Scientist | AI/ML Engineer

syamchand.chilaka@gmail.com | +1 7324768923 | Jersey City, New Jersey | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Professional Summary

AI / ML Engineer with 5+ years shipping production Generative AI and machine learning systems in financial services and consulting. At Citi, built an LLM chatbot on Hugging Face and LangChain that cut customer response times by 80%, and RAG pipelines that lifted document classification accuracy by 50% on financial text. At Accenture, deployed an XGBoost anomaly detection model at 96% accuracy that reduced false positives by 40% in risk monitoring, and an NLP recommender that lifted click through rate by 20% across SaaS campaigns. Now targeting senior AI / ML engineering roles centered on LLMs, RAG, and applied Generative AI.

Skills

- **Generative AI & NLP:** LLMs (BERT, GPT-3, LLaMA), Retrieval-Augmented Generation (RAG), LangChain, Hugging Face, Transformers, Vector Databases, Embeddings
- **Machine Learning:** XGBoost, Recommender Systems, Anomaly & Fraud Detection, Reinforcement Learning, A/B Testing, Deep Learning (CNNs, RNNs, LSTMs), Computer Vision
- **Frameworks:** PyTorch, TensorFlow, Keras, OpenCV
- **Languages:** Python, R, Java, SQL, Scala, Bash, Git
- **MLOps & Deployment:** MLflow, Airflow, AWS SageMaker, Docker, Kubernetes, Jenkins, CI/CD, FastAPI, Model Monitoring, Drift Detection
- **Cloud & Big Data:** AWS (S3, RDS, EC2, Lambda, Bedrock, SageMaker), Azure, GCP BigQuery, Spark (PySpark, MLlib), Databricks, Delta Lake, dbt
- **Visualization & BI:** Tableau, Power BI, Looker

Work Experience

Citi - Data Scientist | New Jersey | Sep 2023 – Present

- Designed and deployed a Generative AI chatbot on Hugging Face Transformers and LangChain, cutting customer response times by 80%.
- Built retrieval-augmented generation (RAG) pipelines with vector databases and feature engineering on financial text, improving document classification accuracy by 50%.
- Automated model retraining and data preparation workflows with MLflow and Airflow, reducing turnaround time by 20%.
- Designed production monitoring dashboards in Tableau and CloudWatch, tracking drift, latency, and uptime of AI models in production.
- Partnered with compliance teams to embed explainability frameworks into LLM solutions, supporting audit and governance reviews.
- Optimized SQL queries and maintained data pipelines, reducing data preparation time for ML models by 20%.
- Presented ML strategy and results to senior leadership, informing prioritization of AI infrastructure investment.

Accenture - Data Scientist II | India | Sep 2020 – Jul 2022

- Built and deployed an XGBoost anomaly detection model at 96% accuracy as an API microservice, reducing false positives by 40% in financial risk monitoring.
- Created an NLP based recommendation engine using embeddings and feature engineering, lifting click through rate by 20% for SaaS marketing campaigns.
- Applied KMeans clustering to identify distinct customer segments, enhancing targeted marketing campaigns by 25%.
- Accelerated distributed training by 30% on Databricks with PyTorch and Spark, and automated ML workflows on AWS SageMaker with MLflow tracking and Jenkins based CI/CD pipelines for reproducibility.
- Embedded explainability frameworks with compliance teams reducing model review cycles by 25%, and mentored 3 junior data scientists on MLOps and feature engineering.

Accenture - Junior Data Scientist | India | Aug 2019 – Aug 2020

- Engineered predictive models (Decision Trees, Random Forests, XGBoost) with advanced feature engineering, achieving 72% accuracy in customer response prediction.
- Developed a computer vision defect detection model (CNNs) deployed at the edge with 95% accuracy.
- Designed and executed A/B testing frameworks in Python, improving conversions by 20%, and automated SQL and Python ETL pipelines cutting ML data preparation time by 20%.
- Automated CI/CD pipelines with Git and Jenkins on Azure (Docker and Kubernetes), reducing build and deployment times for big data projects by 15%.

Projects

AI-Powered Customer Support Chatbot (LLM + RAG) | Citi

- LangChain and Hugging Face LLM chatbot for financial queries; integrated RAG pipelines with vector databases to improve document classification accuracy by 50%; automated retraining workflows via MLflow and Airflow, cutting turnaround time by 20%.

End-to-End ML Platform for Risk & Recommendations | Accenture

- NLP recommendation engine (embeddings + XGBoost) lifting CTR by 20% for SaaS client campaigns; XGBoost anomaly detection at 96% accuracy reducing false positives by 40% in financial risk monitoring; CNN defect classifier deployed at the edge on Docker and Kubernetes (Azure) at 95% accuracy.

Software Engineering Fellow - Project Helix | Handshake AI

- Contributed to training and evaluation of LLM coding assistants by authoring high-quality code samples, reviewing model-generated solutions across Python, SQL, and ML workflows, and providing structured feedback to improve code generation, reasoning, and debugging behavior.

Project Axis - AI Red Teaming and Evaluation | Handshake AI x Stanford HAI

- Task Author on a Stanford HAI research initiative evaluating frontier AI models; designed PhD-level reasoning questions grounded in recent arXiv papers to expose traceable failure modes, built gold-standard solutions with step-by-step calculations, and diagnosed reasoning errors across multiple frontier models through an automated evaluation pipeline; work supported AI safety in healthcare, legal, hiring, and education applications.

Education

Master of Science in Computer and Information Science | **Long Island University, Brooklyn, New York**
Bachelor of Technology in Computer Science | **REVA University, Bangalore, Karnataka, India**

Certifications

- Model Validation 1 (Trainer)
- Model Validation 2 (Expert)
- Microsoft Certified: Azure Fundamentals
- Object Oriented Programming in Java